

COMPUTER VISION

Complete Exam Preparation Guide — 200+ Q&A

M.EIC / M.IA — FEUP

1 — IMAGES AND FILTERING

Fundamentals

Q: What is the effect of adding the same value to all three R,G,B components?

ANSWER:

The hue and saturation are unchanged. Only brightness increases.

Geometrically: moves every pixel along the achromatic axis $(0,0,0) \rightarrow (255,255,255)$ — purely a brightness shift.

Q: What happens if you multiply all three RGB components by the same scalar?

ANSWER:

Contrast increases or decreases relative to zero. If scalar >1 and values stay below max, image brightens and contrast increases proportionally.

Hue and saturation remain unchanged (ratios preserved). If scalar <1 , image darkens.

Q: How do you convert an RGB image to grayscale?

ANSWER:

Weighted sum: $Y = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B$ (luminance-preserving, matches human perception — eyes are most sensitive to green, least to blue).

Simple average $(R+G+B)/3$ ignores perceptual weights and looks visually wrong.

Q: What is the difference between intensity and luminance?

ANSWER:

Intensity: simple arithmetic mean of R,G,B — treats all channels equally.

Luminance: perceptually weighted average (e.g. $Y=0.299R+0.587G+0.114B$) — matches human visual sensitivity to different wavelengths.

Q: What does subtracting the mean from an image do?

ANSWER:

Centers the intensity distribution around zero. Used in preprocessing (zero-mean normalisation) to improve gradient-based learning and comparison.

Does not change contrast (spread) — standard deviation unchanged. Pairs with dividing by std for full standardisation.

Colour Spaces

Q: What are the advantages of HSV over RGB for segmentation?

ANSWER:

RGB couples colour and brightness — a lighting change alters all three channels.

HSV separates: H (hue/actual colour), S (saturation), V (brightness). Allows thresholding on H+S to isolate a colour regardless of illumination. Used widely for colour-based object segmentation.

Q: When would you use the Lab colour space?

ANSWER:

Lab separates luminance (L) from chrominance (a=green-red axis, b=blue-yellow axis). Perceptually uniform: equal distances in Lab correspond to equal perceived differences.

Useful for: colour-based histogram equalisation (equalise L only), colour transfer between images, perceptual colour matching.

Q: What is the YCbCr colour space and where is it used?

ANSWER:

Y = luma (brightness), Cb = blue-difference chroma, Cr = red-difference chroma.

Used in JPEG compression and video coding — humans are more sensitive to luma than chroma, so chroma channels are downsampled (4:2:0) for compression with minimal perceived quality loss.

Q: Why might colour information be insufficient for distinguishing objects?

ANSWER:

Objects of different classes can share the same colour. Also, the same object can appear very different under different lighting conditions or from different viewpoints.

Texture, shape, and context are often more discriminative than colour alone.

Histogram Operations

Q: What is the expected result of histogram equalisation on a grayscale image?

ANSWER:

Redistributes intensities to approximate a uniform histogram, spreading values across the full range.

Effect: increases global contrast, especially when histogram is concentrated in a narrow range.

Formally: applies the CDF of the input as a mapping function: $out = (CDF(in) - CDF_min) / (total_pixels - CDF_min) * (max_val)$.

Q: Why should you NOT equalise R, G, B channels independently?

ANSWER:

Independent equalisation stretches each channel differently, changing the ratios between channels.

This causes colour shifts and artefacts (e.g. originally grey areas become coloured). Correct approach: convert to HSV/Lab, equalise V or L only, convert back.

Q: An image has median intensity 200. After histogram equalisation, brighter or darker?

ANSWER:

Darker. Median=200 means ~50% of pixels have intensity ≤ 200 (78% of 255). The CDF rises steeply at high values.

Equalisation maps intensities to flatten the CDF. The dense cluster of bright pixels gets compressed/mapped lower. Result: overall image becomes darker.

Q: What is CLAHE and when is it preferred over global histogram equalisation?

ANSWER:

CLAHE = Contrast Limited Adaptive Histogram Equalisation. Applies HE locally to small tiles, with a clip limit on contrast enhancement to prevent noise amplification.

Preferred when: global HE over-enhances noise or creates unnatural results in regions of uniform intensity. Common in medical imaging (X-rays, fundus images) where local contrast matters.

Q: What is a histogram backprojection and how is it used?

ANSWER:

Given a reference histogram (e.g. of a target object's colour), compute for each pixel the probability that it belongs to the same distribution.

Result: probability map that highlights pixels likely to be part of the object. Used in object tracking (CamShift algorithm), colour-based segmentation.

Filtering

Q: Why are Gaussian filters preferable to mean (box) filters?

ANSWER:

Mean filter: all pixels in kernel contribute equally — causes ringing/blocking.

Gaussian: weights decrease smoothly from centre — better model of natural blur. Separable (2 x 1D passes → faster). No negative side-lobes → true low-pass. Rotationally symmetric.

Q: What is a median filter and when is it preferred?

ANSWER:

Replaces each pixel with the median of its neighbourhood. Non-linear — cannot be expressed as a convolution.

Preferred for salt-and-pepper (impulse) noise because it completely removes isolated extreme values. Gaussian would spread them. Preserves edges better than Gaussian for this noise type.

Q: What is the difference between correlation and convolution?

ANSWER:

Correlation: slides kernel over image, computes dot product at each position. Kernel is not flipped.

Convolution: same but kernel is flipped 180° before sliding. For symmetric kernels (Gaussian, box) they are equivalent. For asymmetric kernels (Sobel), they differ. Deep learning 'convolutions' are technically correlations.

Q: What does a Laplacian filter detect?

ANSWER:

Second-order derivative of image intensity. Detects regions of rapid intensity change — used for edge detection and sharpening.

Isotropic (detects edges in all directions). Sensitive to noise — usually applied after Gaussian smoothing (Laplacian of Gaussian, LoG).

Zero-crossings of LoG correspond to edge locations.

Q: What is the difference between low-pass and high-pass filtering?

ANSWER:

Low-pass: passes low spatial frequencies, attenuates high. Effect: blurring/smoothing. Examples: Gaussian, mean filter.

High-pass: passes high frequencies, attenuates low. Effect: sharpening, edge enhancement. Examples: Laplacian, unsharp masking (original - blurred).

Q: What is unsharp masking?

ANSWER:

Sharpening technique: $\text{sharpened} = \text{original} + \lambda(\text{original} - \text{blurred})$. The term $(\text{original} - \text{blurred})$ is the high-frequency component (edges, details).

Amplifies fine detail and edges. λ controls strength. Widely used in photography and print.

Q: Explain the concept of a separable filter and its computational advantage.

ANSWER:

A 2D filter is separable if it can be written as outer product of two 1D filters. Example: Gaussian $G(x,y) = G(x) \cdot G(y)$.

Advantage: instead of $O(k^2)$ multiplications per pixel (k =kernel size), apply two 1D passes: $O(2k)$. For $k=5$: 25 ops vs 10. For $k=15$: 225 vs 30. Significant speedup.

Q: What happens in the frequency domain when you apply a Gaussian filter?

ANSWER:

In frequency domain, convolution becomes multiplication. Gaussian in spatial domain is also a Gaussian in frequency domain.

Multiplying by a Gaussian in frequency domain attenuates high frequencies smoothly. No sharp cutoff → no ringing. This is why Gaussian is a better low-pass filter than a box (which has sinc frequency response with side-lobes).

Edge Detection

Q: Describe the Sobel operator.

ANSWER:

Two 3×3 kernels: G_x detects horizontal edges (change in x), G_y detects vertical edges. Gradient magnitude = $\sqrt{G_x^2 + G_y^2}$. Direction = $\text{atan2}(G_y, G_x)$.

Approximates first derivative. Includes Gaussian-like smoothing (weighted central difference). Less noise-sensitive than simple difference filter.

Q: Describe the steps of the Canny edge detector.

ANSWER:

1. Gaussian smoothing. 2. Compute gradient magnitude and direction (Sobel). 3. Non-maximum suppression (thin edges — keep only local maxima along gradient direction). 4. Double thresholding (strong/weak/discard). 5. Hysteresis: keep weak edges connected to strong.

Result: thin, connected, accurate edges. Widely considered optimal edge detector.

Q: What is the aperture problem in edge detection and motion?

ANSWER:

A filter or optical flow sensor can only observe the component of motion/gradient perpendicular to an edge. The component along the edge is undetectable.

Example: for a long horizontal edge moving diagonally, only the vertical motion component is visible through a small aperture. Lucas-Kanade solves this by combining observations from multiple pixels in a patch.

2 — FEATURES / LOCAL FEATURES / WARPING

Harris Corner Detector

Q: How does the Harris corner detector work?

ANSWER:

Computes structure tensor $M = \text{sum over window of } [I_x^2, I_x I_y; I_x I_y, I_y^2]$. Analyses eigenvalues λ_1, λ_2 of M .

Corner: both large. Edge: one large, one small. Flat: both small.

Response $R = \det(M) - k \cdot \text{trace}(M)^2$. If $R > \text{threshold} \rightarrow \text{corner}$.

Q: How can you change Harris sensitivity to detect more/fewer corners?

ANSWER:

Lower threshold $\rightarrow$ more corners. Raise threshold $\rightarrow$ fewer.

Higher k parameter $\rightarrow$ fewer (more selective). Larger Gaussian window $\rightarrow$ more smoothing, may merge corners.

Q: Match eigenvalue pairs: (0,0), (4,4), (0.1,5) to image regions.

ANSWER:

(0,0) $\rightarrow$ flat/uniform. No gradient in any direction.

(4,4) $\rightarrow$ corner. Large gradients in both directions.

(0.1,5) $\rightarrow$ edge. Large gradient perpendicular to edge, small along it.

Q: Why are Harris corners good for Lucas-Kanade tracking?

ANSWER:

Harris matrix M is exactly the matrix used in LK. At corners M is well-conditioned (both eigenvalues large) $\rightarrow$ the linear system has a unique, numerically stable solution. At edges/flat regions, M is ill-conditioned $\rightarrow$ ambiguous or unreliable flow.

Q: What is the Harris-Laplace detector?

ANSWER:

Extends Harris to be scale-invariant by detecting Harris corners at multiple scales using a Gaussian scale-space pyramid. Scale is selected as the one that maximises the normalised Laplacian response.

SIFT, ORB, and Descriptors

Q: How does SIFT achieve scale and rotation invariance?

ANSWER:

Scale: detect keypoints in Difference-of-Gaussians (DoG) pyramid — local extrema in scale-space.

Rotation: compute dominant gradient orientation in local neighbourhood, rotate descriptor to align to 0° .

Descriptor: 128-dim vector (4x4 grid of 8-bin gradient histograms). L2-normalised for illumination invariance.

Q: Do SIFT/ORB operate on grayscale or colour?

ANSWER:

Grayscale. Keypoint detection uses intensity gradients — colour adds no meaningful gradient information. Multiple channels would create separate keypoints per channel with no advantage.

Q: How does ORB differ from SIFT?

ANSWER:

ORB = Oriented FAST + Rotated BRIEF. Detector: FAST (efficient corner detector). Descriptor: BRIEF (binary string — compares pixel pairs), rotated for invariance. Binary descriptor → compared with Hamming distance (XOR + popcount), very fast. Patent-free (SIFT was patented). Less accurate for large viewpoint changes but real-time capable.

Q: What is Lowe's ratio test?

ANSWER:

For each descriptor, find 2 nearest neighbours. Accept match only if $\text{dist_1st}/\text{dist_2nd} < 0.8$. Good match: 1st NN is much closer than 2nd. Ambiguous if both are similar distances → reject. Eliminates ambiguous matches, dramatically reducing false positives.

Q: What is BRIEF and why is it fast?

ANSWER:

Binary Robust Independent Elementary Features. Descriptor = binary string of comparisons between random pixel pairs in a smoothed patch. 256-bit string typical. Extremely fast to compute (pixel comparisons) and match (Hamming distance via XOR). No gradient computation needed.

Q: What is the difference between a keypoint and a descriptor?

ANSWER:

Keypoint (interest point): location (x,y) plus scale, orientation, strength — describes WHERE something is interesting. Descriptor: vector encoding the appearance around that keypoint — describes WHAT it looks like, used for matching across images.

Homographies and Warping

Q: Explain affine vs perspective transformation.

ANSWER:

Affine (6 DOF): preserves parallel lines. Handles translation/rotation/scale/shear. Needs 3 point correspondences. Perspective/Homography (8 DOF): parallel lines can converge. Models full projective transformation. Needs 4 points. Division by w encodes projection plane tilt. Affine is special case where $h_{20}=h_{21}=0$.

Q: What is forward vs inverse warping?

ANSWER:

Forward warping: for each source pixel, compute where it maps to in output. Problem: may leave holes (multiple sources $\rightarrow$ same output, some outputs have none).
Inverse warping: for each OUTPUT pixel, compute where it came from in input, then sample. No holes. Standard approach. Requires interpolation (bilinear/bicubic) for non-integer sample locations.

Q: What is bilinear interpolation?

ANSWER:

When sampling at a non-integer position (x,y) , bilinearly interpolate using the 4 surrounding integer-position pixels: weighted average based on fractional distances.
Linear interpolation first horizontally, then vertically. Produces smooth results, avoids aliasing of nearest-neighbour. Bicubic uses 16 neighbours and is even smoother.

Q: Explain the role of RANSAC in homography estimation.

ANSWER:

Feature matching produces outliers (wrong matches). RANSAC: repeatedly (1) sample 4 random correspondences, (2) compute H , (3) count inliers (reprojection error $<$ threshold). Keep best model. Refit using all inliers.
Robust to outliers. Works if any 4-point sample is all-inliers. N iterations chosen by expected outlier ratio and desired confidence.

Q: What is image mosaicing/stitching and how is it done?

ANSWER:

Combine multiple overlapping images into a single panorama.
Steps: (1) detect and match keypoints across adjacent images, (2) RANSAC to estimate homography between each pair, (3) warp all images to a common reference plane, (4) blend overlapping regions (alpha blending, multi-band blending).

Q: What is the difference between a homography and an essential/fundamental matrix?

ANSWER:

Homography H (3×3): maps 2D points in one image to 2D points in another. Valid when scene is planar or camera purely rotates.
Fundamental matrix F (3×3 , rank 2): encodes epipolar geometry between two views. Maps point in image 1 to epipolar line in image 2. Requires no planar assumption — general 3D scenes.

Q: What is the epipolar constraint?

ANSWER:

Given corresponding point x in image 1, its match x' in image 2 must lie on a specific line (epipolar line): $x'^T F x = 0$. Reduces correspondence search from 2D to 1D.
The epipolar lines all pass through the epipole (projection of one camera centre into the other camera).

3 — SEGMENTATION / MOTION

Thresholding and Classical Segmentation

Q: Give an example where a single global threshold fails.

ANSWER:

Uneven illumination: document photographed with light source on one side. Background on bright side is brighter than text on dark side. No single threshold correctly separates text from background everywhere.

Fix: adaptive/local thresholding — threshold varies per neighbourhood (e.g. local mean minus constant C).

Q: What is Otsu's method?

ANSWER:

Automatic threshold selection that maximises inter-class variance (equivalently minimises intra-class variance) between foreground and background.

Assumes bimodal histogram (two distinct pixel groups). Optimal for images with clear foreground/background separation. Fails on multi-modal histograms.

Q: How does watershed segmentation work and what is its main drawback?

ANSWER:

Treats gradient magnitude as topographic surface. Simulates flooding from local minima. Where floods from different basins meet = watershed line = boundary.

Drawback: over-segmentation — noise creates too many local minima → too many tiny regions. Fix: marker-controlled watershed (user-defined seeds).

Q: Explain GrabCut.

ANSWER:

User provides bounding box. Pixels outside = background. Iterates: (1) fit GMMs to foreground/background, (2) graph cut assigns pixels to GMMs by minimising energy (data term + smoothness term), (3) repeat until convergence.

Can refine with user strokes. More accurate than thresholding.

Q: What is region growing segmentation?

ANSWER:

Start from seed point(s). Iteratively add neighbouring pixels to a region if they satisfy a similarity criterion (e.g. intensity within threshold of region mean).

Simple but sensitive to seed placement and threshold. Can leak through weak boundaries.

Q: What is the mean shift algorithm?

ANSWER:

Iterative mode-finding in feature space (colour, position). Each pixel moves toward nearest mode (density peak) of the feature distribution. Pixels converging to same mode → same segment.

Does not require number of clusters. Robust to irregular shapes. Slow for large images.

Deep Learning Segmentation

Q: Why do DL methods outperform traditional methods for semantic segmentation?

ANSWER:

Traditional methods use hand-crafted features — no notion of semantics. DL learns hierarchical representations: edges → parts → semantic concepts. Generalises better via large-dataset training. DL also jointly learns features and decision boundaries, whereas traditional methods separate these steps.

Q: What is an FCN (Fully Convolutional Network)?

ANSWER:

Replaces FC layers of a classification CNN with conv layers, producing spatial output maps instead of a single class score.

Uses transposed convolutions (learned upsampling) to restore spatial resolution. Skip connections from earlier layers add back fine spatial detail. Pioneered end-to-end trainable semantic segmentation.

Q: Explain U-Net architecture and why it works for segmentation.

ANSWER:

Encoder-decoder with skip connections. Encoder: downsample (conv+pool), increasing depth. Decoder: upsample (transposed conv) restoring resolution. Skip connections: concatenate encoder feature maps to decoder.

Encoder captures context (what), decoder recovers location (where), skip connections preserve fine boundaries. Originally designed for biomedical images with limited training data.

Q: What is DeepLab and what problem does it solve?

ANSWER:

DeepLab series introduced atrous (dilated) convolutions to increase receptive field without reducing resolution or adding parameters.

Also introduced Atrous Spatial Pyramid Pooling (ASPP): parallel atrous convolutions at multiple dilation rates to capture multi-scale context. v3+ adds encoder-decoder structure.

Q: What are dilated (atrous) convolutions?

ANSWER:

Standard conv with gaps between filter elements, controlled by dilation rate r . A 3×3 filter with dilation=2 covers a 5×5 area. Increases receptive field exponentially without losing resolution or adding parameters.

Used in DeepLab, WaveNet. Avoids pooling which discards spatial information.

Q: What is instance vs semantic vs panoptic segmentation?

ANSWER:

Semantic: classify every pixel into a category (all people = 'person').

Instance: detect and mask individual object instances (person 1, person 2, ...) — only 'things', not 'stuff'.

Panoptic: every pixel has class label AND instance ID. Unifies both: things get unique instance IDs, stuff gets semantic label only.

Q: What is Mask R-CNN and how does it perform instance segmentation?

ANSWER:

Extends Faster R-CNN with third head: small FCN predicts binary pixel mask per RoI in parallel with class/box heads.

RoI Align (vs RoI Pool): bilinear interpolation preserves spatial precision for better masks. Multi-class: class head identifies object type, mask head produces the contour.

Optical Flow and Motion

Q: Comment: 'Optical flow is not always a correct estimate of motion.'

ANSWER:

Optical flow tracks brightness patterns, not physical surfaces. They diverge when brightness constancy assumption fails.

Classic example: uniform-colour rotating ball — no brightness change → zero optical flow despite real motion. Also: moving light source makes stationary object appear to move.

Q: Explain Lucas-Kanade optical flow.

ANSWER:

Assumes brightness constancy: $I(x,y,t) = I(x+u,y+v,t+1)$. Taylor expansion: $I_x \cdot u + I_y \cdot v + I_t = 0$. One equation, two unknowns (aperture problem).

Fix: assume constant flow in window (25 equations, 2 unknowns). Least squares: $M[u;v] = -b$, where M is Harris matrix. Well-conditioned at corners. Fails for large motion → fix with image pyramids.

Q: What is the Horn-Schunck method vs Lucas-Kanade?

ANSWER:

Horn-Schunck: global method. Minimises global energy = sum of brightness constancy error + smoothness term (penalises flow discontinuities). Produces dense, smooth flow field.

Lucas-Kanade: local method. Fits flow in each patch independently. Less smooth but handles motion boundaries better (doesn't force smoothness across object edges). More commonly used.

Q: What is dense optical flow vs sparse optical flow?

ANSWER:

Sparse: compute flow only at selected keypoints (e.g. Harris corners with LK). Fast, used for tracking.

Dense: compute flow at every pixel (e.g. Farneback, RAFT, FlowNet). Slow but complete motion field. Used for video understanding, frame interpolation.

Q: For each optical flow pattern, describe scene/camera: radial outward, uniform rightward, rotational.

ANSWER:

Radial outward from centre: camera moving forward (dolly/zoom into scene) — focus of expansion.

Uniform rightward: camera panning horizontally, or objects all moving uniformly right.

Rotational: camera spinning around optical axis, or rotating object seen head-on.

Q: What is background subtraction and how does it work?

ANSWER:

Separates moving foreground from static background in video.

Simple: frame difference (current - previous frame, threshold). Better: maintain running model of background (Gaussian per pixel, or MOG2 — Mixture of Gaussians per pixel). Foreground = pixels that deviate significantly from background model.

Handles slow lighting changes by updating background model adaptively.

Q: What is the difference between motion detection and motion estimation?

ANSWER:

Motion detection: binary — 'is there motion here?' (background subtraction, frame differencing).

Motion estimation: quantitative — 'how much and in which direction?' (optical flow gives velocity vector per pixel). Motion estimation is harder and more informative.

4 — CONVOLUTIONAL NEURAL NETWORKS

Architecture and Operations

Q: What is learned in early vs late CNN layers?

ANSWER:

Early: low-level generic features — oriented edges, colour gradients, textures. Highly transferable.
Mid: parts, shapes, texture combinations. Late conv: task-specific abstract representations. FC layers: classification decision boundary.

Q: 'It is possible to replace FC layers with another classifier (e.g. SVM).' Comment.

ANSWER:

True. CNN without FC acts as feature extractor → SVM (or any classifier) on top. Basis of original R-CNN.

Advantage: needs less labelled data. Disadvantage: not end-to-end — features not optimised for the task. Fine-tuning the whole network (when enough data) outperforms fixed features + SVM.

Q: Five 3x3 filters, stride 2, no padding, applied to 11x11x30 input. Output shape?

ANSWER:

Spatial: $\text{floor}((11-3)/2)+1 = 5$. Depth = number of filters = 5. Output: 5×5×5.
Input depth (30) is consumed per filter — each filter spans full input depth.

Q: How do you count parameters in a conv layer?

ANSWER:

$(\text{filter_H} * \text{filter_W} * \text{input_depth} + 1) * \text{num_filters}$. +1 for bias per filter.
Example: 3x3 conv, 64 filters, 32 input channels: $(9*32+1)*64 = 18,496$.
Independent of input spatial resolution.

Q: Does changing input resolution from 224x224 to 128x128 change number of trainable parameters?

ANSWER:

No for conv layers — same filters, same weights. Yes for FC layers — they depend on flattened spatial size.
This is why architectures with GlobalAvgPool (no FC layers after feature extraction) are resolution-agnostic.

Q: What is the role of pooling layers?

ANSWER:

Reduce spatial dimensions → fewer parameters in subsequent layers. Provide translation invariance (small shifts don't change output). Increase effective receptive field.
Max Pool: keep most prominent activation. Average Pool: smooth mean. GlobalAvgPool: average entire feature map to scalar → 0 parameters.

Q: If GlobalAvgPool is replaced with Flatten+FC, what happens to parameter count?

ANSWER:

Significantly more parameters. GlobalAvgPool produces a vector of length=num_feature_maps (0 parameters). Flatten produces a large vector ($H*W*C$) and the FC layer has $(H*W*C)*output_size$ parameters — much larger.

Q: What is stride and how does it differ from pooling?

ANSWER:

Stride > 1: skip positions during convolution — reduces spatial output size while computing features. Also increases receptive field.

Pooling: separate layer applied after conv. Both reduce spatial size but stride in conv reduces it during feature computation, preserving more information. Modern architectures often prefer strided conv over pooling.

Q: What is padding and what are 'same' vs 'valid' padding?

ANSWER:

Valid: no padding → output smaller than input.

Same: zero-pad input so output has same spatial size as input (with stride=1). Formula: $pad=(filter-1)/2$ for odd filter size.

Same padding preserves spatial dimensions across layers, making architecture design simpler.

Q: What is a receptive field and why does it matter?

ANSWER:

Region of input image that influences a neuron's activation. Grows with depth: after k conv layers of size f , $RF=(k*(f-1)+1)$.

Too small: model only sees local texture, cannot model global structure. Object recognition needs RF covering the whole object. Pooling and strided conv increase RF growth rate.

Training

Q: What is the vanishing gradient problem and how does it affect deep networks?

ANSWER:

In deep networks, gradients of early layers become very small (product of many values <1 via chain rule) → early layers learn very slowly.

Solutions: ReLU activations (no saturation for positive values), batch normalisation, residual connections (skip connections provide gradient highway), proper weight initialisation (Xavier, He).

Q: What is batch normalisation and why is it useful?

ANSWER:

Normalises layer activations to zero mean and unit variance per mini-batch, then applies learnable scale (γ) and shift (β).

Benefits: reduces internal covariate shift, allows higher learning rates, reduces sensitivity to weight initialisation, mild regularisation effect. Placed after conv/FC, before activation.

Q: What is dropout and how does it regularise a network?

ANSWER:

Randomly set a fraction p of activations to zero during training. Forces network to learn redundant representations (cannot rely on any single neuron).

At test time: scale activations by $(1-p)$ or use inverted dropout (scale during training). Prevents co-adaptation of neurons. Effective regulariser especially for FC layers.

Q: What is the difference between L1 and L2 regularisation?

ANSWER:

L2 (weight decay): adds $\lambda \sum(w^2)$ to loss. Penalises large weights, pushes all weights toward zero uniformly. Leads to small distributed weights.

L1: adds $\lambda \sum(|w|)$. Promotes sparsity — many weights exactly zero. Used for feature selection.

Q: What is data augmentation and why is it important?

ANSWER:

Artificially expand training set by applying label-preserving transformations: random crop, flip, rotation, colour jitter, cutout, Mixup, CutMix.

Reduces overfitting, improves generalisation, especially when labelled data is scarce. Essential in medical imaging.

Q: Explain transfer learning: feature extraction vs fine-tuning.

ANSWER:

Feature extraction: freeze pre-trained weights, train only new head. Use when: small dataset, similar domain.

Fine-tuning: unfreeze some/all layers, continue training with small LR. Use when: larger dataset or domain shift. Rule: small+similar → freeze more. Large+different → fine-tune more.

Q: What is the difference between underfitting and overfitting?

ANSWER:

Underfitting: model too simple — high training AND test error. Fix: larger model, more features, less regularisation, train longer.

Overfitting: model memorises training data — low training error, high test error. Fix: more data, augmentation, dropout, regularisation, early stopping, simpler model.

Q: What activation functions are commonly used and what are their trade-offs?

ANSWER:

Sigmoid: outputs $(0,1)$, saturates → vanishing gradient. Used in output layer for binary classification.

ReLU: $\max(0,x)$. No vanishing gradient for positives. Dead neurons (always 0 if input always negative). Default for hidden layers.

LeakyReLU: small negative slope → no dead neurons. GELU: smooth approximation of ReLU, used in Transformers.

Architectures

Q: What are residual connections (ResNet) and what problem do they solve?

ANSWER:

Skip connections add input to output of a block: $y = F(x) + x$. Network learns residual $F(x) = \text{desired_output} - x$ rather than full mapping.

Solves vanishing gradient: gradient flows directly through skip connections. Enables training of very deep networks (100+ layers). State-of-the-art backbone.

Q: What are depthwise separable convolutions (MobileNet)?

ANSWER:

Split standard conv into: (1) depthwise conv — one filter per channel (spatial patterns), (2) pointwise conv — 1×1 across channels.

~8-9x fewer parameters and operations than standard conv. Used in MobileNet, EfficientNet, Xception for efficient mobile inference.

Q: What is an Inception module?

ANSWER:

Parallel branches with different filter sizes (1×1 , 3×3 , 5×5) and max pooling, all applied to same input and concatenated along depth. Captures multi-scale spatial features simultaneously.

1×1 convolutions used for dimensionality reduction (bottleneck) before expensive $3 \times 3/5 \times 5$ ops.

Q: What is the bottleneck block in ResNet?

ANSWER:

3-layer block: 1×1 conv (reduce channels), 3×3 conv (spatial features), 1×1 conv (restore channels). More efficient than simple 3×3 blocks for deep networks.

Used in ResNet-50 and deeper. 1×1 convs reduce computational cost of the 3×3 central conv.

5 — OBJECT DETECTION / SEGMENTATION WITH CNNs

Two-Stage Detectors

Q: Why is R-CNN slow?

ANSWER:

~2000 region proposals via selective search. Each region independently warped and forward-passed through full CNN → ~2000 CNN passes per image. Selective search itself also slow.

Q: How did Fast R-CNN improve over R-CNN?

ANSWER:

Run CNN once on full image to get shared feature map. Extract RoI features using RoI Pooling (no per-region CNN forward pass). Train classification+regression jointly end-to-end.
Main remaining bottleneck: selective search for region proposals (runs on CPU, ~2 sec/image).

Q: Why is Faster R-CNN ~10x faster than R-CNN?

ANSWER:

(1) CNN backbone runs ONCE on full image → shared feature map. (2) RPN replaces selective search — learns proposals from feature map. (3) RoI Pooling extracts fixed features per proposal without re-running CNN.

Q: What is the Region Proposal Network (RPN)?

ANSWER:

Small network (3x3 conv + two 1x1 conv heads) sliding over feature map. At each location, predicts objectness score and box offsets for k anchor boxes (different scales/ratios).
~300 proposals (filtered from ~20k anchors) passed to detector head. Trained jointly with detector.

Q: What is RoI Pooling vs RoI Align?

ANSWER:

RoI Pooling: quantises RoI boundaries to integer coordinates, then max pools each bin. Quantisation error degrades localisation (~1-2 pixels shift).

RoI Align: uses bilinear interpolation at exact floating-point locations within each bin. No quantisation → significantly better mask accuracy (crucial for Mask R-CNN).

Single-Stage Detectors

Q: Explain YOLO's architecture and why 'You Only Look Once' is apt.

ANSWER:

Image divided into SxS grid. Single CNN forward pass outputs $S \times S \times (B \times 5 + C)$ tensor — bounding boxes, confidences, class probs all at once. No proposal stage.

Name is apt: exactly one forward pass on full image for complete detection. Contrast with Faster R-CNN's two-stage pipeline.

Q: YOLO v1: 10 classes, 7x7 grid, 2 boxes per cell. How many output values?

ANSWER:

$7 * 7 * (2*5 + 10) = 7*7*20 = 980$ values. Each cell: 2 boxes * (x,y,w,h,conf) + 10 class probs.

Q: What problem does focal loss solve and how?

ANSWER:

Class imbalance in dense detectors: massive background anchors dominate loss. Model spends effort on easy negatives.

Focal loss = $-(1-pt)^\gamma * \log(pt)$. Down-weights easy examples (high pt → small factor). Focus training on hard misclassified examples. Enabled RetinaNet to match two-stage accuracy.

Q: What is SSD (Single Shot Detector)?

ANSWER:

Applies detection heads at multiple feature map scales (from different layers of backbone) simultaneously. Different scales detect different object sizes.

No RPN, no FC layers. Faster than two-stage, worse on very small objects. Uses default/anchor boxes.

Segmentation Architectures

Q: What CNN method for segmenting individual blood cells of different types?

ANSWER:

Mask R-CNN: instance segmentation. Detects each cell individually (RPN), classifies its type (multi-class head), produces pixel mask (FCN head per RoI). Handles overlapping instances. RoI Align for precise masks.

Q: Explain U-Net and why it works for both segmentation and image generation.

ANSWER:

Encoder-decoder + skip connections. Encoder: semantic context. Decoder: spatial reconstruction. Skips: preserve fine details.

For generation (diffusion): encode noisy input to extract structure, decode to generate, skips preserve detail — same architecture maps image → image naturally.

Q: What is panoptic segmentation?

ANSWER:

Every pixel assigned class label AND instance ID. 'Things' (countable objects) get unique IDs. 'Stuff' (sky, road) gets only semantic label.

Unifies semantic + instance segmentation. Evaluated by panoptic quality (PQ).

Evaluation Metrics

Q: What is IoU and how is it used?

ANSWER:

Intersection over Union: $\text{area}(A \cap B) / \text{area}(A \cup B)$. Ranges 0–1. Standard detection quality metric. Used in NMS (suppress boxes with $\text{IoU} > \text{threshold}$), and as TP criterion ($\text{IoU} > 0.5$ typically counts as correct detection).

Q: Explain Non-Maximum Suppression (NMS).

ANSWER:

Sort detections by confidence. Take highest-confidence box, keep it. Suppress remaining boxes with $\text{IoU} > \text{threshold}$ vs kept box. Repeat.

Removes redundant detections per object. Soft-NMS: instead of removing, down-weights by overlap — better for overlapping objects.

Q: What is mAP (mean Average Precision)?

ANSWER:

For each class: sort detections by confidence, compute precision-recall curve, integrate → AP (area under PR curve). mAP = mean over all classes.

COCO mAP averages over IoU thresholds 0.5:0.05:0.95. More rigorous than single-threshold mAP.

Q: What is precision vs recall in object detection?

ANSWER:

Precision = $TP / (TP + FP)$ — of all detected boxes, fraction are correct.

Recall = $TP / (TP + FN)$ — of all ground truth objects, fraction were detected.

High precision: few false alarms. High recall: few missed detections. Trade-off controlled by confidence threshold.

6 — GENERATIVE ADVERSARIAL NETWORKS

Fundamentals

Q: What is the GAN training objective?

ANSWER:

Min-max game: G minimises $\log(1-D(G(z)))$, D maximises $\log(D(x))+\log(1-D(G(z)))$.
Equivalently G maximises $\log(D(G(z)))$ (non-saturating version — better gradients early in training).
Nash equilibrium: G produces perfect samples, D outputs 0.5 everywhere.

Q: What is mode collapse in GANs?

ANSWER:

Generator maps many different noise inputs to same (or few) output mode(s). Ignores input diversity.
D easily fooled by the few modes G learned.
Fixes: minibatch discrimination, WGAN, unrolled GANs, diverse training objectives.

Q: What is Wasserstein GAN and how does it improve stability?

ANSWER:

Uses Wasserstein-1 (Earth Mover) distance instead of JS divergence. Provides stable gradients even when distributions don't overlap (JS divergence saturates).
Requires weight clipping (original) or gradient penalty (WGAN-GP) to enforce Lipschitz constraint on D (now called 'critic').

Q: What is a Conditional GAN?

ANSWER:

Both $G(z,c)$ and $D(x,c)$ receive conditioning info c (class label, text, image). Enables controlled generation of specific classes or attributes.
Applications: Pix2Pix (paired image-to-image), class-conditional generation, text-to-image.

Q: Could a GAN address class imbalance? How?

ANSWER:

Yes — train conditional GAN on minority class, generate synthetic samples to balance dataset.
GAN augmentation better than oversampling (no new info) or simple augmentation (limited variation).
GAN learns true distribution → genuinely diverse new samples.
Caveat: needs enough minority samples to train GAN. Very rare classes: combine with transfer learning.

Specialised GAN Types

Q: Why is cycle consistency loss important in CycleGAN?

ANSWER:

Unpaired translation (e.g. horses → zebras). Without it: G can map any horse to same realistic zebra (mode collapse), ignoring horse content.
Cycle consistency: $F(G(x)) \approx x$. G must preserve content for F to reconstruct. Prevents mode collapse, enforces correspondence.

Q: What is Pix2Pix?

ANSWER:

Supervised image-to-image translation with paired training data. Conditional GAN + L1 loss (ensures correspondence with ground truth). PatchGAN discriminator: classifies $N \times N$ patches as real/fake rather than whole image.

Applications: sketch → photo, aerial → map, day → night, edges → image.

Q: What is a Progressive GAN?

ANSWER:

Train GAN starting at low resolution (4x4), progressively add layers to grow to higher resolution (up to 1024x1024).

Stabilises training: easy low-res structure learned first, then fine details. Enabled high-quality face generation (CelebA-HQ).

Q: What is StyleGAN?

ANSWER:

Extends Progressive GAN with style-based generator. Noise → mapping network (MLP) → intermediate latent W . Style vectors injected via AdaIN at each resolution level.

Disentangled latent space: W controls global structure, local textures separately. Enables style mixing, interpolation.

Q: What is an image-to-image translation task? Give examples.

ANSWER:

Transform images from one domain to another. Paired: Pix2Pix (sketches → photos). Unpaired: CycleGAN (horses → zebras, summer → winter).

Applications: photo enhancement, medical image synthesis (MRI → CT), style transfer, data augmentation.

7 — ATTENTION MODELS / VISION TRANSFORMERS

Self-Attention

Q: Explain the self-attention mechanism (Q, K, V).

ANSWER:

$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) * V$. Q=query (what am I looking for), K=key (what do I contain), V=value (what I contribute).

For each query: dot-product with all keys → softmax → attention weights → weighted sum of values.
Sqrt(d_k) scaling prevents saturation.

Q: What is multi-head attention?

ANSWER:

Run h parallel attention heads with different Q/K/V projections (each dimension $d_k = d_{\text{model}}/h$).
Concatenate outputs, apply linear projection.

Each head can attend to different aspects (spatial position, semantics, colour). More expressive than single-head.

Q: What is the complexity of self-attention and why is it a concern?

ANSWER:

$O(n^2 * d)$ for n tokens, d dimensions — quadratic in sequence length. For high-resolution images, n is large (e.g. $224 \times 224 / 16 \times 16 = 196$ tokens, manageable; $512 \times 512 / 4 \times 4 = 16384$ tokens, expensive).
Solutions: windowed attention (Swin), linear attention approximations, sparse attention patterns.

Vision Transformer (ViT)

Q: How does ViT process an image?

ANSWER:

(1) Split into $N \times N$ patches (typically 16×16). (2) Flatten each patch, linear projection to embedding dim. (3) Add positional encodings. (4) Prepend [CLS] token. (5) Transformer encoder (MSA+MLP blocks). (6) [CLS] output for classification.

Q: A cat's eye patch attends strongly to ears/whiskers but weakly to nearby background. What does this suggest?

ANSWER:

Model learned semantic relationships over spatial proximity. Builds representations based on 'same object' not 'nearby pixels'.

Beneficial vs CNN: global context in single layer, data-driven receptive field, better small object understanding, handles occlusion via distant semantic cues.

Q: Is it easier to combine image and text with CNN or Transformer?

ANSWER:

Transformer — much easier. Both modalities become token sequences. Concatenate, apply same self-attention mechanism.

CNN: spatial grid incompatible with variable-length text sequence. Must manually design fusion (FiLM,

concatenation layer, bolted-on cross-attention) — extra engineering.

Q: What are the advantages and disadvantages of ViT vs CNN?

ANSWER:

ViT advantages: global context per layer, naturally scales with data, modality-agnostic, better on large datasets.

ViT disadvantages: needs large datasets (no spatial locality bias), higher compute, worse than CNNs on small datasets.

CNN advantages: data-efficient (inductive biases), faster at small scale.

Q: What is Swin Transformer?

ANSWER:

Hierarchical ViT using shifted window attention. Attention computed within local windows (not globally) → linear complexity. Windows shift between layers to allow cross-window information flow.

Builds hierarchical feature maps (like CNN) → suitable for dense prediction tasks (detection, segmentation). State-of-the-art on many benchmarks.

Q: What is DINO?

ANSWER:

Self-supervised ViT training. Student network processes augmented crops of image, teacher (EMA of student) processes full image. Student trained to match teacher's [CLS] token output.

Emergent property: DINO attention heads learn semantic segmentation without supervision — just from the self-supervised objective.

Q: How is cross-attention used for conditioning in diffusion models?

ANSWER:

Noisy latent features → Query. Conditioning embeddings (e.g. CLIP text) → Keys and Values. Every spatial position attends to relevant condition tokens.

Injected at intermediate U-Net resolutions. Spatial feature for 'dog' region attends strongly to 'dog' text token.

8 — DIFFUSION MODELS

Q: What are diffusion models and how do they generate images?

ANSWER:

Forward process: add Gaussian noise over T steps until pure noise. Fixed, no learning.

Reverse process: network (U-Net) learns to predict noise at each step, iteratively denoising from pure noise to image.

Training: minimise $\|\epsilon - \epsilon_\theta(x_t, t)\|^2$ — predict the noise added at timestep t .

Q: Why run diffusion in latent space (Stable Diffusion)?

ANSWER:

Pixel-space: $512 \times 512 \times 3 \approx 786K$ dims. Hundreds of denoising steps in this space → very expensive.

Solution: pretrained VAE encodes to compact latent (e.g. $64 \times 64 \times 4$). Diffusion runs in $\sim 16x$ smaller space. Decoder maps latent to pixels once at end.

Q: Explain classifier-free guidance.

ANSWER:

Single model trained conditional ($\sim 90\%$) and unconditional ($\sim 10\%$, null condition).

Inference: $\epsilon_{\text{hat}} = \epsilon(x, \emptyset) + w[\epsilon(x, c) - \epsilon(x, \emptyset)]$. Pushes prediction toward conditional minus unconditional.

$w=1$: no guidance. $w=7.5$: typical default. $w>10$: over-saturated. Controls diversity vs fidelity tradeoff.

Q: How does ControlNet work?

ANSWER:

Frozen original U-Net + trainable copy of encoder+middle. Spatial conditioning (edges/pose/depth) fed to trainable copy.

Zero convolutions (init to zero) connect copy to frozen decoder. At start: zero output → no harm to pretrained weights. Gradually learns to steer generation. Modality-agnostic.

Q: What is DDPM vs DDIM?

ANSWER:

DDPM (Denoising Diffusion Probabilistic Models): stochastic reverse process, needs $T=1000$ steps for quality.

DDIM (Denoising Diffusion Implicit Models): deterministic reverse process using same trained model. Can skip timesteps → 50 steps instead of 1000, $\sim 20x$ faster, minimal quality loss.

Q: What is the role of the VAE in Stable Diffusion?

ANSWER:

Encoder E : compresses pixel image to latent z (perceptual compression). Decoder D : reconstructs pixel image from latent.

Pretrained separately on image reconstruction. During diffusion training/inference, E and D are frozen. Enables diffusion to work in a smaller, semantically rich space.

Q: What is score matching in the context of diffusion models?

ANSWER:

The noise prediction network ϵ_θ learns to predict the score (gradient of log probability) of the data distribution, which is related to the noise at each step.

Score-based models and DDPM are mathematically equivalent frameworks — both learn to reverse a diffusion process via noise estimation.

Q: What types of conditioning are possible in Stable Diffusion?

ANSWER:

Text prompt (via CLIP text encoder + cross-attention), image (via CLIP image encoder or IP-Adapter), semantic map, depth map, pose skeleton (ControlNet), inpainting mask, class label.

Conditioned via cross-attention in U-Net layers or by concatenation with latent.

9 — CAMERA MODELS & 3D VISION

Q: What is the pinhole camera model?

ANSWER:

Projects 3D world point (X,Y,Z) to 2D image point (x,y) via: $x=f*X/Z$, $y=f*Y/Z$, where f is focal length. In homogeneous coordinates: $[u,v,1]^T \propto K * [R|t] * [X,Y,Z,1]^T$. K = intrinsic matrix (focal length, principal point). $[R|t]$ = extrinsic (camera pose).

Q: What are camera intrinsics vs extrinsics?

ANSWER:

Intrinsics K : internal parameters — focal length (f_x, f_y) , principal point (c_x, c_y) , skew. Fixed for a given camera and lens.

Extrinsics $[R|t]$: position and orientation of camera in world coordinates. Change with camera placement.

Q: What is lens distortion and how is it corrected?

ANSWER:

Real lenses have radial distortion (barrel/pincushion) and tangential distortion. Straight lines appear curved.

Correction: estimate distortion coefficients during calibration (using checkerboard pattern), apply inverse distortion map to undistort image before processing.

Q: What is stereo vision and how does it estimate depth?

ANSWER:

Two cameras with known baseline b . For a point at depth Z , disparity $d = b*f/Z$ (difference in horizontal position between left and right images).

Steps: rectify images (align epipolar lines horizontally), compute disparity map (match corresponding pixels), compute depth from disparity.

Q: What is camera calibration?

ANSWER:

Estimate intrinsic (K) and distortion parameters from images of a known calibration target (e.g. checkerboard).

Zhang's method: observe multiple checkerboard poses, solve for K and distortion using correspondences between checkerboard corners and their known 3D positions.

Q: What is structure from motion (SfM)?

ANSWER:

Reconstruct 3D structure of scene and camera poses from multiple unordered images.

Steps: feature detection+matching across images, essential/fundamental matrix estimation, triangulation to get 3D points, bundle adjustment (global optimisation of cameras+structure simultaneously).

Q: What is SLAM and how does it relate to CV?

ANSWER:

Simultaneous Localisation and Mapping. Robot/device builds a map of environment while tracking its own location within it.

Visual SLAM uses camera as sensor (ORB-SLAM, LSD-SLAM). Key components: feature tracking/optical flow, keyframe selection, loop closure detection, pose graph optimisation.

10 — SELF-SUPERVISED & REPRESENTATION LEARNING

Q: What is self-supervised learning in computer vision?

ANSWER:

Learning visual representations from unlabelled images using automatically generated supervisory signals (pretext tasks) derived from the data itself.

Benefits: leverage vast unlabelled data. Learned representations transfer well to downstream tasks with limited labels.

Q: What are contrastive learning methods (SimCLR, MoCo)?

ANSWER:

Learn representations by: pull together embeddings of different augmented views of same image (positive pairs), push apart embeddings from different images (negative pairs).

SimCLR: large batch, in-batch negatives, projection head during training removed at fine-tuning.

MoCo: momentum encoder + memory queue as negatives, avoids needing huge batches.

Q: What is CLIP and how does it work?

ANSWER:

Trains image encoder and text encoder jointly on 400M (image, text) pairs from the internet.

Objective: contrastive — matching image-text pairs should have high similarity, non-matching pairs low.

Zero-shot capability: classify images by computing similarity to text descriptions of classes without any task-specific training.

Q: What is masked image modelling (MAE)?

ANSWER:

Masked AutoEncoder: mask random patches (75%) of image, train ViT to reconstruct them. Forces model to learn semantic understanding.

Similar to BERT for text. Efficient: encoder only processes visible patches (not masked). Learns strong representations for downstream tasks.

Q: What is knowledge distillation?

ANSWER:

Train a smaller student network to mimic a larger teacher network. Student trained to match teacher's soft output probabilities (not just hard labels).

Soft labels encode inter-class similarities (e.g. '90% cat, 8% dog') — more informative than one-hot.

Enables compact models with near-teacher accuracy.

Q: What is few-shot learning?

ANSWER:

Learn to classify new classes from very few examples (1-shot, 5-shot). Key idea: meta-learning — train across many tasks so model learns to learn quickly.

Prototypical Networks: embed few-shot examples, compute class prototypes (mean embedding), classify by nearest prototype.

11 — VIDEO UNDERSTANDING

Q: How are 3D convolutions used for video?

ANSWER:

Extend 2D spatial conv to 3D (space+time). Filter size: $T \times H \times W$. Captures both spatial and temporal patterns jointly.

Expensive: C3D, I3D. Decomposed: (2+1)D convolutions (spatial 2D + temporal 1D) — similar accuracy, fewer parameters, easier optimisation.

Q: What is temporal pooling and why is it needed?

ANSWER:

Videos have variable length. Temporal pooling aggregates frame-level features into a single video representation.

Options: average pooling, max pooling, learned pooling (NetVLAD, transformer over frames). Choice affects sensitivity to temporal order.

Q: What is the two-stream architecture for action recognition?

ANSWER:

Parallel streams: (1) spatial stream — single RGB frame for appearance, (2) temporal stream — stacked optical flow frames for motion.

Late fusion combines predictions. Spatial captures who/what, temporal captures how they move. Two-stream outperforms either alone.

Q: What is video object detection and what makes it harder than image detection?

ANSWER:

Detect objects in every frame of a video. Additional challenges: motion blur, defocus blur, occlusion across frames, computational efficiency.

Solutions: feature aggregation across frames (FGFA), tracking-by-detection, propagate detections using optical flow.

12 — ADVANCED DETECTION & RECOGNITION

Q: What is Feature Pyramid Network (FPN)?

ANSWER:

Builds multi-scale feature pyramid with top-down pathway + lateral connections. Deepest layer (most semantic, least resolution) upsampled and added to shallower layers.

Detects objects at each scale level. Combines semantic richness of deep layers with spatial precision of shallow layers.

Q: What is anchor-free detection?

ANSWER:

Detects objects without predefined anchor boxes. Predicts object centre and size directly.

Examples: CornerNet (detect box corners as keypoints), CenterNet (object = Gaussian heatmap at centre point), FCOS (per-pixel prediction of offsets to box edges).

Avoids hyperparameter tuning of anchor scales/ratios.

Q: What makes DETR architecturally different?

ANSWER:

Fully end-to-end: CNN features → Transformer encoder → N object queries through Transformer decoder → N predictions simultaneously.

No anchors, no NMS, no RPN. Hungarian matching loss. Advantage: elegant pipeline. Disadvantage: slow convergence, poor on small objects.

Q: What is Spatial Pyramid Pooling (SPP)?

ANSWER:

Pool feature map at multiple scales (e.g. 4×4, 2×2, 1×1), concatenate all pooled outputs → fixed-length vector regardless of input resolution.

Solves fixed-input-size limitation of FC layers. Used in SPPNet and YOLOv3+.

Q: What are deformable convolutions?

ANSWER:

Learn additional offset fields per spatial location, shifting sampling positions adaptively. Network learns where to look — adapts receptive field to object geometry.

Useful for deformable/rotating objects. Used in Deformable DETR.

Q: What is knowledge in Siamese networks?

ANSWER:

Two (or more) weight-sharing subnetworks process different inputs, produce embeddings. Distance metric compares them.

Applications: face verification, signature verification, one-shot learning, visual tracking (match template to patches).

Trained with contrastive or triplet loss.

13 — EVALUATION, METRICS & PRACTICAL

Q: What metrics are used for image classification?

ANSWER:

Top-1 accuracy: fraction of images where highest-confidence prediction is correct. Top-5 accuracy: correct class in top 5 predictions.

F1 score for imbalanced classes: harmonic mean of precision and recall.

Q: What metrics for semantic segmentation?

ANSWER:

Pixel accuracy: fraction of correctly classified pixels. Mean IoU (mIoU): average IoU across all classes — standard metric. Weighted IoU: weighted by class frequency.

mIoU is preferred because pixel accuracy is misleading when class frequencies are very unbalanced.

Q: What metrics for instance segmentation?

ANSWER:

AP at various IoU thresholds (mask AP). COCO benchmark uses AP averaged over IoU 0.5:0.05:0.95.

Mask AP evaluates both detection accuracy (finding the instance) and segmentation quality (pixel-level mask).

Q: What is the bias-variance tradeoff?

ANSWER:

Bias: error from wrong model assumptions. High bias = underfitting (model too simple).

Variance: sensitivity to training set fluctuations. High variance = overfitting (model too complex).

Increasing model complexity decreases bias, increases variance. Regularisation, ensembling, and more data help manage the tradeoff.

Q: What is the difference between validation set and test set?

ANSWER:

Validation set: used during development to tune hyperparameters and select models. Can influence model indirectly.

Test set: held out completely until final evaluation. Should be used exactly once. Multiple evaluations on test set inflate results.

Q: How would you handle class imbalance in a dataset?

ANSWER:

Data-level: oversample minority (SMOTE, GAN augmentation), undersample majority.

Algorithm-level: class-weighted loss, focal loss.

Evaluation: use precision/recall/F1 and per-class metrics instead of overall accuracy.

Q: What is cross-validation and when is it used?

ANSWER:

K-fold: split data into k subsets, train on $k-1$, validate on 1, repeat k times, average results. Gives better estimate of generalisation than single train/val split.

Used when dataset is small. Expensive but reliable.

14 — IMAGE GENERATION & SYNTHESIS

Q: What is a Variational Autoencoder (VAE)?

ANSWER:

Encoder maps image to distribution (μ , σ) in latent space. Sample $z \sim N(\mu, \sigma^2)$. Decoder reconstructs image from z .

Loss = reconstruction loss + KL divergence (forces latent to be close to standard normal). KL term makes latent space continuous/structured → enables sampling new images.

Q: How does a VAE differ from a standard autoencoder?

ANSWER:

Standard AE: deterministic latent → not generative (cannot sample new images by sampling latent). Latent space may have gaps.

VAE: stochastic latent — encoder outputs distribution, sample from it. Continuous structured latent space via KL regularisation → can sample and interpolate between images.

Q: What is neural style transfer?

ANSWER:

Optimise a new image to simultaneously match: content (high-level feature activation) of a content image + style (Gram matrices of feature maps) of a style image.

Gram matrix captures feature correlations (texture statistics) at each layer. Content loss at deep layer (semantic), style loss at multiple layers (texture at multiple scales).

Q: What are Gram matrices in neural style transfer?

ANSWER:

Gram matrix $G = F * F^T$ where F is the feature map (channels $\times$ spatial). Each element G_{ij} = correlation between feature maps i and j .

Captures texture/style statistics (which features co-occur) without regard to spatial location. Matched between generated and style images to transfer texture without copying structure.

Q: What is image inpainting?

ANSWER:

Fill in missing or masked regions of an image coherently. Classic: PatchMatch (copy-paste similar patches from elsewhere in image). DL: partial convolutions (mask-aware), LaMa (Fourier convolutions), diffusion-based inpainting.

Diffusion models are state-of-the-art: condition on masked image + noise prediction within masked region.

Q: What is super-resolution?

ANSWER:

Upscale a low-resolution image to higher resolution, hallucinating plausible fine details.

Classic: bicubic interpolation (blurry). DL: SRCNN, SRGAN (adds perceptual + adversarial loss for sharp realistic textures), ESRGAN, diffusion-based SR.

Loss: MSE → blurry results (favours average). Perceptual loss + GAN loss → sharper, more realistic.

15 — NUMERIC PROBLEMS BANK

Q: General formula for conv output shape.

ANSWER:

$\text{floor}((\text{input} + 2 * \text{padding} - \text{filter}) / \text{stride}) + 1$. Depth = num_filters.

Same padding: output = $\text{ceil}(\text{input} / \text{stride})$. Valid padding: output = $\text{floor}((\text{input} - \text{filter}) / \text{stride}) + 1$.

Q: 32x32x3 input, 5x5 conv, 16 filters, stride 1, padding 2. Output shape and parameter count?

ANSWER:

Spatial: $\text{floor}((32 + 4 - 5) / 1) + 1 = 32$. Output: 32x32x16.

Parameters: $(5 * 5 * 3 + 1) * 16 = 1,216$.

Q: 224x224x3 input through AlexNet-like stack: conv(11,96,s4,p0) → pool(3,s2) → conv(5,256,s1,p2). Shape after each?

ANSWER:

After conv1: $\text{floor}((224 - 11) / 4) + 1 = 54 \rightarrow 54 \times 54 \times 96$.

After pool1: $\text{floor}((54 - 3) / 2) + 1 = 26 \rightarrow 26 \times 26 \times 96$.

After conv2 (same padding): $26 \times 26 \times 256$.

Q: YOLO v1: S=7, B=2, C=20. Total output values?

ANSWER:

$7 * 7 * (2 * 5 + 20) = 7 * 7 * 30 = 1,470$.

Q: How many parameters in a fully connected layer mapping 512 inputs to 10 outputs (with bias)?

ANSWER:

$512 * 10 + 10 = 5,130$. (weight matrix + bias vector)

Q: A 3-layer network: FC(512 → 256), FC(256 → 128), FC(128 → 10). Total parameters (with bias)?

ANSWER:

Layer 1: $512 * 256 + 256 = 131,328$. Layer 2: $256 * 128 + 128 = 32,896$. Layer 3: $128 * 10 + 10 = 1,290$.

Total: 165,514.

Q: What happens to output shape when you apply GlobalAvgPool to a 7x7x512 feature map?

ANSWER:

Output: 512 (one scalar per feature map, spatial dimensions collapsed to 1x1). No trainable parameters.

Q: Depthwise separable vs standard conv: 3x3, 64 in channels, 128 out channels. Parameter ratio?

ANSWER:

Standard: $(9 * 64 + 1) * 128 = 73,856$.

Depthwise sep: $(9 \cdot 64 + 64) + (1 \cdot 64 \cdot 128 + 128) = 640 + 8,320 = 8,960$.

Ratio: $73856/8960 \approx 8.2\times$ fewer parameters.

16 — NICHE / TRICKY TOPICS

Q: What is the lottery ticket hypothesis?

ANSWER:

Dense networks contain sparse subnetworks ('winning tickets') that, when trained from their original initialisation, reach similar accuracy as the full network.

Implication: random initialisation matters — the 'winning ticket' subnetwork can be found by magnitude pruning + rewinding weights to original init.

Q: What is quantisation in neural networks?

ANSWER:

Reduce precision of weights/activations (e.g. float32 → int8). Reduces model size (4x) and speeds up inference (integer arithmetic faster on hardware).

Post-training quantisation: quantise after training, possible accuracy drop. Quantisation-aware training: simulate quantisation during training for better accuracy.

Q: What is pruning?

ANSWER:

Remove unnecessary weights/neurons/filters from trained network. Magnitude pruning: zero out smallest weights. Structured pruning: remove entire filters (hardware-friendly).

Followed by fine-tuning to recover accuracy. Reduces model size and inference cost.

Q: What is the difference between semantic and instance features in detection?

ANSWER:

Semantic features: class-specific (what is it?). High-level, deep in network.

Instance features: location-specific (where exactly?). Need fine spatial precision. Preserved by skip connections / FPN. Both are needed for detection+segmentation.

Q: What is label smoothing?

ANSWER:

Instead of hard one-hot targets (0 or 1), use softened targets: (1-epsilon) for true class, epsilon/(C-1) for others.

Prevents model from being over-confident. Acts as regularisation. Improves calibration of probability outputs.

Q: What is test-time augmentation (TTA)?

ANSWER:

Apply multiple augmentations (flip, crop, rotation) to test image, average predictions. Improves accuracy at cost of inference time.

Model effectively 'sees' multiple views of same image — especially useful for competition leaderboards.

Q: Why do we use stride 2 in conv instead of stride 1 + pooling in modern architectures?

ANSWER:

Strided conv learns the downsampling — it can preserve relevant information while reducing resolution.

Pooling always uses a fixed operation (max/avg). Learned strided conv potentially more expressive. Modern architectures (ResNet, EfficientNet) increasingly prefer strided conv.

Q: What is the inductive bias of a CNN vs ViT?

ANSWER:

CNN: translation equivariance (shifting input shifts feature map equally), locality (nearby pixels more related than distant ones). Helps with limited data.

ViT: no such bias — must learn spatial relationships from data. Needs more data to match CNN but potentially learns better representations at large scale.

Q: What is feature map vs activation map?

ANSWER:

Often used interchangeably. Technically: feature map = output tensor of a conv layer (result of applying one filter). Activation map = feature map after applying activation function (ReLU etc.).

Q: What is a 1x1 convolution and what is it used for?

ANSWER:

Applies a learned linear combination across channels at each spatial position. No spatial context.

Uses: dimensionality reduction/expansion (bottleneck layers), channel mixing (Inception, MobileNet pointwise), adding non-linearity between channels without spatial op.

Q: What is the difference between object detection and object recognition?

ANSWER:

Recognition: 'is there a cat in this image?' — image-level label, no localisation.

Detection: 'where are the cats?' — bounding boxes + class labels. Detection requires both classification and localisation.

Q: Why do we normalise image pixel values before feeding to a CNN?

ANSWER:

Neural network gradient descent works better when inputs have zero mean and unit variance. Raw pixels [0-255] have wrong scale.

Standard practice: subtract ImageNet mean [0.485, 0.456, 0.406] and divide by std [0.229, 0.224, 0.225] per channel. This ensures gradients have consistent scale.

Q: What is class activation mapping (CAM)?

ANSWER:

Visualises which regions of an image activated a particular class prediction. Computed as weighted sum of final conv feature maps, where weights are those of the global average pooled + FC classifier.

Requires GlobalAvgPool before classifier. GradCAM generalises to any layer using gradients.

Q: What is GradCAM?

ANSWER:

Gradient-weighted Class Activation Mapping. Computes gradients of target class score w.r.t. feature maps of a chosen conv layer. Average gradients spatially → importance weights per feature map → weighted sum → heatmap.

Works on any CNN architecture (no GAP required). Used for model interpretability/debugging.

QUICK REFERENCE — KEY FORMULAS & FACTS

Formula/Fact	Value/Expression
Conv output size	$\text{floor}((\text{in} + 2p - k) / s) + 1$
Conv parameters	$(k \cdot k \cdot C_{\text{in}} + 1) \cdot C_{\text{out}}$
Harris response	$R = \det(M) - k \cdot \text{trace}(M)^2$
YOLO output tensor	$S \times S \times (B \cdot 5 + C)$
IoU	$ A \cap B / A \cup B $
Attention	$\text{softmax}(QK^T / \sqrt{d_k}) \cdot V$
Brightness constancy (LK)	$I_x \cdot u + I_y \cdot v + I_t = 0$
Focal Loss	$-(1 - p_t)^\gamma \cdot \log(p_t)$
CFG formula	$\mathcal{E}(x, c) = \mathcal{E}(x, \emptyset) + w \cdot [\mathcal{E}(x, c) - \mathcal{E}(x, \emptyset)]$
Cycle loss	$F(G(x)) \approx x \text{ AND } G(F(y)) \approx y$
Grayscale from RGB	$Y = 0.299R + 0.587G + 0.114B$
Stereo depth	$Z = b \cdot f / \text{disparity}$

Model	Type	Key Feature
R-CNN	2-stage det.	Selective search + per-region CNN
Fast R-CNN	2-stage det.	Shared feature map + RoI Pool
Faster R-CNN	2-stage det.	RPN replaces selective search
YOLO	1-stage det.	Single pass, grid regression
SSD	1-stage det.	Multi-scale heads
RetinaNet	1-stage det.	Focal loss
Mask R-CNN	Instance seg	+ mask head (FCN per RoI)
U-Net	Sem. seg	Encoder-decoder + skip connections
DeepLab v3+	Sem. seg	Dilated conv + ASPP
DETR	Detection	Transformer, no anchors, no NMS
ViT	Classification	Patch embeddings + self-attention
Swin	Hier. ViT	Shifted window attention
CycleGAN	Image trans.	Unpaired + cycle consistency
Stable Diffusion	Generation	Latent diffusion + cross-attn
ControlNet	Controlled gen	Frozen backbone + zero-conv branch
CLIP	Multimodal	Image-text contrastive pretraining